We create chemistry

Safety data sheet

BASF safety data sheet. This is a translation of the country-specific safety data sheet into a language other than that required by law. This document does not replace the safety data sheet provided according to Regulation (EC) No 1907/2006.

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Product: **Styrene 500 BMB I**

(ID no. 30738050/SDS_GEN_DE/EN)

Date of print 09.07.2025

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1. Product identifier

Styrene 500 BMB I

Chemical name: STYRENE 500
CAS Number: 100-42-5

1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Monomer for manufacturing of polymers
Uses advised against: All consumer uses are strongly advised against.
Recommended use: Monomer., for industrial processing only

For the detailed identified uses of the product see appendix of the safety data sheet.

1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Telephone: +49 621 60-0
E-mail address: global.info@basf.com

1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

SECTION 2: Hazards Identification

2.1. Classification of the substance or mixture

According to Regulation (EC) No 1272/2008 [CLP]

Asp. Tox. 1	H304 May be fatal if swallowed and enters airways.
Flam. Liq. 3	H226 Flammable liquid and vapour.
Acute Tox. 4 (Inhalation - vapour)	H332 Harmful if inhaled.
Skin Corr./Irrit. 2	H315 Causes skin irritation.
Eye Dam./Irrit. 2	H319 Causes serious eye irritation.
Repr. 2	H361d Suspected of damaging the unborn child.
STOT SE 3	H335 May cause respiratory irritation.
STOT RE 1	H372 Causes damage to organs (Auditory organ) through prolonged or repeated exposure.
Aquatic Chronic 3	H412 Harmful to aquatic life with long lasting effects.

For the classifications not written out in full in this section the full text can be found in section 16.

2.2. Label elements

According to Regulation (EC) No 1272/2008 [CLP]

Pictogram:

Signal Word:

Danger

Hazard Statement:

H226	Flammable liquid and vapour.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H304	May be fatal if swallowed and enters airways.
H335	May cause respiratory irritation.
H361d	Suspected of damaging the unborn child.
H372	Causes damage to organs (Auditory organ) through prolonged or repeated exposure.
H412	Harmful to aquatic life with long lasting effects.

Precautionary Statements (Prevention):

P271	Use only outdoors or in a well-ventilated area.
P280	Wear protective gloves, protective clothing and eye protection or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260	Do not breathe dust/gas/mist/vapours.
P273	Avoid release to the environment.

P243	Take action to prevent static discharges.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.
P241	Use explosion-proof electrical, ventilating and lighting equipment.
P270	Do not eat, drink or smoke when using this product.
P264	Wash contaminated body parts thoroughly after handling.
P240	Ground and bond container and receiving equipment.
P242	Use non-sparking tools.

Precautionary Statements (Response):

P308 + P311	IF exposed or concerned: Call a POISON CENTER or physician.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P314	Get medical advice/attention if you feel unwell.
P301 + P310	IF SWALLOWED: Immediately call a POISON CENTER or physician.
P303 + P361 + P353	IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.
P332 + P313	If skin irritation occurs: Get medical attention.
P331	Do NOT induce vomiting.
P337 + P311	If eye irritation persists: Call a POISON CENTER or physician.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P370 + P378	In case of fire: Use water spray, dry powder, foam or carbon dioxide for extinction.

Precautionary Statements (Storage):

P233	Keep container tightly closed.
P403 + P235	Store in a well-ventilated place. Keep cool.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

2.3. Other hazards

According to Regulation (EC) No 1272/2008 [CLP]

See section 12 - Results of PBT and vPvB assessment.
Product does not contain a substance above legal limits included in the list established in accordance with Article 59(1) of Regulation (EC) No 1907/2006 for having endocrine disrupting properties or is identified to have endocrine disrupting properties in accordance with the criteria set out in Commission Delegated Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605.

SECTION 3: Composition/Information on Ingredients

3.1. Substances

Chemical nature

styrene

Content (W/W): 99 % - 100 %	Asp. Tox. 1
CAS Number: 100-42-5	Flam. Liq. 3
EC-Number: 202-851-5	Acute Tox. 4 (Inhalation - vapour)
	Skin Corr./Irrit. 2
	Eye Dam./Irrit. 2
	Repr. 2 (unborn child)
	STOT SE 3 (irr. to respiratory syst.)
	STOT RE (Auditory organ) 1
	Aquatic Chronic 3
	H226, H319, H315, H332, H304, H335, H361d, H372, H412

stabilized with:
4-tert-Butylpyrocatechol

	Skin Corr./Irrit. 1B
CAS Number: 98-29-3	Eye Dam./Irrit. 1
EC-Number: 202-653-9	Skin Sens. 1
	Aquatic Chronic 2
	H317, H314, H411

Regulatory relevant ingredients

styrene

Content (W/W): >= 99 % - <= 100 %	Asp. Tox. 1
	Flam. Liq. 3
CAS Number: 100-42-5	Acute Tox. 4 (Inhalation - vapour)
EC-Number: 202-851-5	Skin Corr./Irrit. 2
	Eye Dam./Irrit. 2
	Repr. 2 (unborn child)
	STOT SE 3 (irr. to respiratory syst.)
	STOT RE (Auditory organ) 1
	Aquatic Chronic 3
	H226, H319, H315, H332, H304, H335, H361d, H372, H412

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

3.2. Mixtures

Not applicable

SECTION 4: First-Aid Measures

4.1. Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

SECTION 5: Fire-Fighting Measures

5.1. Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
No data available.

5.2. Special hazards arising from the substance or mixture

Advice: Do not breathe gas/vapour.

Advice: Flammable liquid Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

5.3. Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

SECTION 6: Accidental Release Measures

High risk of slipping due to leakage/spillage of product.

Release of substance/product can cause fire or explosion. Shut off or stop source of leak. Shut off or stop released substance/product under safe conditions.

Pack in tightly closed containers for disposal.

6.1. Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame. Use antistatic tools.

6.2. Environmental precautions

Discharge into the environment must be avoided.

6.3. Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.
For large amounts: Dike spillage. Place into suitable container for disposal.
For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

SECTION 7: Handling and Storage

7.1. Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice.

Protection against fire and explosion:
Avoid all sources of ignition: heat, sparks, open flame. Ground all transfer equipment properly to prevent electrostatic discharge.

Electrical devices must meet the specified temperature class.

7.2. Conditions for safe storage, including any incompatibilities

Suitable materials for containers: Carbon steel (Iron), Stainless steel 1.4401, Stainless steel 1.4301 (V2), tinned carbon steel (Tinplate), Aluminium, glass, Galvanized carbon steel (Zinc), Stainless steel 1.4361, Stainless steel 1.4541, Stainless steel 1.4439, Stainless steel 1.4571, Stainless steel 1.4539, Stainless steel 1.4306 (V2A), Stainless steel 1.4311, Stainless steel 1.4404
Unsuitable materials for containers: brass, Copper

Storage class according to TRGS 510 (originally VCI, Germany): (3) Flammable liquids

Storage stability:
Storage temperature: < 25 °C
Check 2-3 times per week to ensure that stabilizer content is adequate.
Storage temperature: > 25 °C
Check daily to ensure that stabilizer content is adequate.

additives:
4-tert-Butylpyrocatechol (CAS Number: 98-29-3)

7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

SECTION 8: Exposure Controls/Personal Protection

8.1. Control parameters

Components with occupational exposure limits

The surveillance of the workplace by exposure measurements may be necessary, in order to prove the efficiency of safety measures, for example ventilation or the need of respiratory protection. Since this requires a specific competency, only accredited laboratories should be contracted. Regarding

suitable methods to assess inhalation exposure, the European Standards EN 482, 689 and 14042 are to be considered. In addition, the TRGS 402 has to be observed in Germany.

100-42-5: styrene

OEL 86 mg/m3 ; 20 ppm (TRGS 900 (DE))
Ceiling limit value/factor: 2
If the occupational exposure limit value (AGW) and the biological limit value (BGW) are complied with, there should be no risk of damage for the unborn child (see TRGS 900, Number 2.7)
Short Term Exposure Classification: (TRGS 900 (DE))
Category II: Substances with a resorptive effect

Biological limit values (BLV)

TRGS 903 (DE)
Determinant: mandelic acid and phenyl glyoxylic acid
Biological Specimen: Creatinine in urine
Sampling time period is for long-term exposures, at the end of the shift after several preceding ones./ Sampling time period is at end of exposure or at end of shift.
Concentration: 600 mg/g Creatinine

PNEC

freshwater: 0,028 mg/l

marine water: 0,0028 mg/l

intermittent release: 0,04 mg/l

sediment (freshwater): 0,614 mg/kg

sediment (marine water): 0,0614 mg/kg

soil: 0,2 mg/kg

STP: 5 mg/l

DNEL

worker:
Long-term exposure- systemic effects, Inhalation: 85 mg/m3

worker:
Long-term exposure- systemic effects, dermal: 406 mg/kg

8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic compounds and solid and liquid particles (f.e. EN 14387 Type A-P2)

Hand protection:
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
fluoroelastomer (FKM) - 0.7 mm coating thickness
Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust)., Low flammability, oil resistant protective clothing (e.g. in accordance with EN 465), safety shoes (f.e. according to EN 20346), antistatic

General safety and hygiene measures

Wearing of closed work clothing is required additionally to the stated personal protection equipment. Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Avoid contact with skin and eyes. At the end of the shift the skin should be cleaned and skin-care agents applied.

Environmental exposure controls

All appropriate measures must be taken to prevent the release of this product to the environment and to limit the dispersion of any release when it occurs. Suitable risk management measures should be in place.

SECTION 9: Physical and Chemical Properties

9.1. Information on basic physical and chemical properties

State of matter:	liquid
Form:	liquid
Colour:	colourless, clear
Odour:	aromatic, sweetish
Odour threshold:	0,05 - 0,08 ppm
Melting point:	-31 °C Literature data.
Boiling point:	145 °C (1.013 hPa) Literature data.

Sublimation point:		
	not applicable	
Flammability:	Flammable.	(derived from flash point)
Flammability of Aerosol Products:		
	not applicable, the product does not form flammable aerosoles	
Lower explosion limit:	1,1 %(V) For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	6,1 %(V) For liquids not relevant for classification and labelling.	
Flash point:	31 °C Literature data.	
Auto-ignition temperature:	490 °C Literature data.	
pH value:		
	The substance does not dissociate.	
Viscosity, dynamic:	0,696 mPa.s (25 °C) Literature data.	
Thixotropy:	not thixotropic	
Solubility in water:	Literature data. 320 mg/l (25 °C)	
Solubility (quantitative) solvent(s):	organic solvents	
Partitioning coefficient n-octanol/water (log Kow):	2,96 (25 °C) The data refers to the undissociated form of the substance.	(OECD Guideline 107)
Vapour pressure:	6,67 hPa (20 °C) Literature data.	(measured)
Relative density:	0,903 - 0,909 (20 °C) Literature data.	
Density:	0,903 - 0,909 g/cm3 (20 °C) Literature data.	
Relative vapour density (air):	> 1 (20 °C) Heavier than air.	(estimated)

9.2. Other information

Information with regard to physical hazard classes

Explosives

Explosion hazard: Based on the chemical structure there is no indication of explosive properties.

Impact sensitivity: not shock-sensitive
Based on the chemical structure there is no shock-sensitivity.

Oxidizing properties

Fire promoting properties: not fire-propagating

Flammable liquids

Sustained combustibility:
not determined

Pyrophoric properties

Self-ignition temperature: Test type: Spontaneous self-ignition at room-temperature.

not self-igniting

Self-heating substances and mixtures

Self heating ability: It is not a substance capable of spontaneous heating.

Substances and mixtures, which emit flammable gases in contact with water

Formation of flammable gases:
Forms no flammable gases in the presence of water.

Corrosion to metals

No corrosive effect on metal.

Other safety characteristics

pKA:
The substance does not dissociate.

Adsorption/water - soil: KOC: 352; log KOC: 2,55 (calculated)

Surface tension:
Based on chemical structure, surface activity is not to be expected.

Molar mass: 104,15 g/mol

SECTION 10: Stability and Reactivity

10.1. Reactivity

When heated can give off ignitable vapours.

Corrosion to metals:	No corrosive effect on metal.	
Formation of flammable gases:	Remarks:	Forms no flammable gases in the presence of water.

10.2. Chemical stability

The product is stable if stored and handled as prescribed/indicated.

10.3. Possibility of hazardous reactions

Radical formation can cause exothermic polymerization. Polymerizes explosively when heated or in contact with bases, acids and metal ions. Risk of spontaneous and violent self-polymerization if inhibitor is lost or product is exposed to excessive heat. Risk of exothermic reaction. Risk of spontaneous polymerization when heated or in the presence of UV radiation. Risk of spontaneous polymerization in the presence of radical donors. Polymerization occurs with acids and acid forming substances.

10.4. Conditions to avoid

Avoid prolonged exposure to extreme heat. Avoid formation of polymers in valves and pipes. Avoid inhibitor loss.

10.5. Incompatible materials

Substances to avoid:
peroxides, oxidizing agents, vinyl polymer catalysts, caustics, copper alloys, metallic halides (salts)

10.6. Hazardous decomposition products

Thermal decomposition products:
aromatic hydrocarbons

SECTION 11: Toxicological Information

11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Assessment of acute toxicity:

Of moderate toxicity after short-term inhalation. High concentrations in the air may cause narcosis. Virtually nontoxic after a single skin contact. Virtually nontoxic after a single ingestion. Harmful: may cause lung damage if swallowed.

Experimental/calculated data:

LD50 rat (oral): 5.000 mg/kg
Literature data.
LC50 rat (by inhalation): 12 mg/l 2770 ppm 4 h
Literature data. The vapour was tested.

LD50 rat (dermal): > 2.000 mg/kg (OECD Guideline 402)

Irritation

Assessment of irritating effects:

Eye contact causes irritation. Skin contact causes irritation.

Experimental/calculated data:

Skin corrosion/irritation

rabbit: Irritant.

Serious eye damage/irritation

rabbit: Irritant.

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies.

Experimental/calculated data:

Guinea pig maximization test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was mutagenic in various test systems with microorganisms and cell cultures; however, these results could not be confirmed in tests with mammals.

Carcinogenicity

Assessment of carcinogenicity:
A clear indication of an increased risk of cancer in humans has so far not been shown. IARC (International Agency for Research on Cancer) has classified this substance as group 2A (The agent is probably carcinogenic to humans).

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:

Possible risk of harm to the unborn child. EU-classification

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Causes temporary irritation of the respiratory tract.

Origin of data: human data

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

The substance may cause deafness after repeated inhalation.

Aspiration hazard

May also damage the lung at swallowing (aspiration hazard).

Interactive effects

No data available.

11.2. Information on other hazards

Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

SECTION 12: Ecological Information

12.1. Toxicity

Assessment of aquatic toxicity:
Acutely toxic for aquatic organisms. Harmful to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) 4,02 mg/l, Pimephales promelas (other, Flow through.)
Literature data.

Aquatic invertebrates:
EC50 (48 h) 4,7 mg/l, Daphnia magna (OECD Guideline 202, part 1, Flow through.)
Literature data.

Aquatic plants:
EC50 (72 h) 4,9 mg/l (growth rate), Selenastrum capricornutum (other)
Literature data.

EC10 (96 h) 0,28 mg/l (growth rate), Desmodesmus subspicatus (other)

Microorganisms/Effect on activated sludge:
EC20 (30 min) 140 mg/l, activated sludge (OECD Guideline 209, aquatic)

Chronic toxicity to fish:
Study does not need to be conducted.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) 1,01 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
The statement of the toxic effect relates to the analytically determined concentration.

Assessment of terrestrial toxicity:
Toxic effects have been observed in studies with soil living organisms.

Soil living organisms:
LC50 (14 d) 120 mg/kg, Eisenia foetida (OECD Guideline 207, artificial soil)

Terrestrial plants:
No data available.

Other terrestrial non-mammals:
No data available.

12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Readily biodegradable (according to OECD criteria).

Elimination information:
70,9 % BOD of the ThOD (28 d) (OECD 301F; ISO 9408; 92/69/EEC, C.4-D) (aerobic, activated sludge, domestic)

Assessment of stability in water:
No data available.
Study scientifically not justified.

12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:

Bioconcentration factor(BCF): 74, Fish (calculated)
Based on a weight of evidence, the compound will not bioaccumulate.

12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will rapidly evaporate into the atmosphere from the water surface.
Adsorption in soil: Adsorption to solid soil phase is not expected.

12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

12.6. Endocrine disrupting properties

The substance is not identified to have endocrine disrupting properties according to Regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 nor is included in the Candidate List of substances of very high concern according to EU REACh Article 59 for having endocrine disrupting properties.

12.7. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

12.8. Additional information

Other ecotoxicological advice:
According to the criteria of Guidelines 67/548/EEC and 1999/45/EC the product is not to classsify as environmental hazard.

SECTION 13: Disposal Considerations

13.1. Waste treatment methods

Dispose of in accordance with national, state and local regulations.

Contaminated packaging:
Disposal must be made according to official regulations.

SECTION 14: Transport Information

Land transport

ADR

UN number or ID number: UN2055

UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	Tunnel code: D/E

RID

UN number or ID number:	UN2055
UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

Inland waterway transport

ADN

UN number or ID number:	UN2055
UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no
Special precautions for user:	None known

Transport in inland waterway vessel

UN number or ID number:	UN2055
UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3, INST, N3
Packing group:	III
Environmental hazards:	yes
Type of inland waterway vessel:	N
Cargo tank design:	3
Cargo tank type:	2

Sea transport

IMDG

UN number or ID number:	UN 2055
UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	no Marine pollutant: NO
Special precautions for user:	EmS: F-E; S-D

Air transport

IATA/ICAO

UN number or ID number:	UN 2055
UN proper shipping name:	STYRENE MONOMER, STABILIZED
Transport hazard class(es):	3
Packing group:	III
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for user:	None known

14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

14.3. Transport hazard class(es)
See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

14.4. Packing group
See corresponding entries for “Packing group” for the respective regulations in the tables above.

14.5. Environmental hazards
See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

14.6. Special precautions for user
See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

14.7. Maritime transport in bulk according to IMO instruments

Regulation:	IBC-Code
Product name:	Styrene monomer
Pollution category:	Y
Ship Type:	3

SECTION 15: Regulatory Information

15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

Prohibitions, Restrictions and Authorizations

Annex XVII of Regulation (EC) No 1907/2006: Number on List: 3, 40, 3, 40, 75

Hazardous Incident Ordinance (Germany):
List entry in regulation: 1.2.5.3

Directive 2012/18/EU - Control of Major Accident Hazards involving dangerous substances (EU):
List entry in regulation: P5c

Classification according to 'TA-Luft' (Germany):

5.2.5 class I: Organic gases class I

Water hazard class (§6 AwSV para.4 (Legal binding announcement of the substance in the Federal Gazette)): (2) significantly water polluting. ID-No.: 187

Regulation on prohibitions and restrictions on the marketing of dangerous substances, preparations and goods in accordance with the chemical law (Germany)
The specifications of the Technical Rule for Hazardous Substances (TRGS) 401 must be observed (TRGS 401: Risks resulting from skin contact - identification, assessment, measures).
German Regulation TA Luft (Technical Instruction on Air Quality Control, i.e. first Directive to the Federal Immission Control Ordinance)
Law on the Protection of Working Youth
The Maternity Protection Act needs to be considered.

15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Acute Tox. 4 (Inhalation - vapour)
Eye Dam./Irrit. 2A
Skin Corr./Irrit. 2
Asp. Tox. 1
Flam. Liq. 3
Aquatic Acute 2
STOT RE (Auditory organ) 1
STOT SE 3 (irritating to respiratory system)
Aquatic Chronic 3
Repr. 2 (unborn child)

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

Asp. Tox.	Aspiration hazard
Flam. Liq.	Flammable liquids
Acute Tox.	Acute toxicity
Skin Corr./Irrit.	Skin corrosion/irritation
Eye Dam./Irrit.	Serious eye damage/eye irritation
Repr.	Reproductive toxicity
STOT SE	Specific target organ toxicity — single exposure
STOT RE	Specific target organ toxicity — repeated exposure
Aquatic Chronic	Hazardous to the aquatic environment - chronic
H226	Flammable liquid and vapour.
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H332	Harmful if inhaled.
H304	May be fatal if swallowed and enters airways.
H335	May cause respiratory irritation.
H361d	Suspected of damaging the unborn child.
H372	Causes damage to organs (Auditory organ) through prolonged or repeated exposure.
H412	Harmful to aquatic life with long lasting effects.

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution

from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.